



ANDERSON JUNIOR COLLEGE

Preliminary Examinations 2011

CHEMISTRY

Higher 1

Paper 1 Multiple Choice

8872/01

22 September 2011

50 minutes

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the Multiple Choice Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

Multiple Choice Answer Sheet

Write your name, PDG and NRIC/FIN number. **DO NOT** include the reference letter.

Shade the NRIC / FIN number.

Exam Title: JC2 Prelim

Exam Details: H1 Chem / Paper 1

Date: 22/09/2011

This document consists of **12** printed pages.

Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1 How many electrons need to be removed to completely ionise 1.0×10^{-6} mol of Rb atoms to Rb^+ ions?

- A** 6.02×10^{23}
B 1.0×10^{-6}
C $1.0 \times 10^{-6} \times 6.02 \times 10^{23}$
D $\frac{1.0 \times 10^{-6}}{6.02 \times 10^{23}}$

- 2 A 25 cm^3 sample of a 0.1 mol dm^{-3} ACl_2 solution requires 10 cm^3 of 0.1 mol dm^{-3} of acidified potassium manganate(VII) solution for complete reaction.

What is the final oxidation state of element **A**?

- A** +3
B +4
C +5
D +6

- 3 0.200 mol of a hydrocarbon undergoes complete combustion to give 26.4 g of carbon dioxide and 10.8 g of water as the only products.

What is the molecular formula of the hydrocarbon?

- A** C_2H_4
B C_2H_6
C C_3H_6
D C_6H_{12}

- 4 In which species are the number of electrons and neutrons equal?

- A** ${}^9_4\text{Be}$ **B** ${}^{35}_{17}\text{Cl}^+$ **C** ${}^{18}_8\text{O}^{2-}$ **D** ${}^{24}_{12}\text{Mg}^{2+}$

- 5 Which of the following reactions results in an increase in the bond angle of the underlined substance?

- A** $\text{Ca}\underline{\text{Si}}\text{O}_3 + 8\text{HF} \longrightarrow \text{H}_2\text{SiF}_6 + \text{CaF}_2 + 3\text{H}_2\text{O}$
B $\text{Na}\underline{\text{N}}\text{H}_2 + \text{H}_2\text{O} \longrightarrow \text{NaOH} + \text{NH}_3$
C $\underline{\text{B}}\text{H}_3 + \text{NaH} \longrightarrow \text{NaBH}_4$
D $\text{H}_2\text{O}_2 + \underline{\text{S}}\text{O}_2 \longrightarrow \text{H}_2\text{SO}_4$

- 6 In which of the following pairs of compounds would the melting point of compound I be higher than compound II?

	Compound I	Compound II
A	KCl	NaCl
B	AlF ₃	AlCl ₃
C	NH ₃	H ₂ O
D	SiCl ₄	SiO ₂

- 7 The co-ordination number of an ion is the number of oppositely charged ions that surround it in an ionic crystal lattice. This number depends mainly on two factors – the relative sizes of the ions and their relative charges. The table below shows the data for the ions in the lattices of NaCl, CsCl and MgO.

compound	cation radius / nm	anion radius / nm	$\frac{\text{anion radius}}{\text{cation radius}}$	co-ordination no. of cation	co-ordination no. of anion
NaCl	0.10	0.18	1.8	6	6
CsCl	0.17	0.18	1.1	8	8
MgO	0.07	0.14	2.0	6	6

Which of the following would be **unlikely** co-ordination numbers for the corresponding compound?

	compound	cation radius / nm	anion radius / nm	co-ordination no. of cation	co-ordination no. of anion
A	NaBr	0.10	0.20	6	6
B	BaO	0.14	0.14	8	8
C	CaF ₂	0.10	0.13	8	4
D	MgCl ₂	0.07	0.18	6	6

- 8 The data below refer to the radii and charges of six ions.

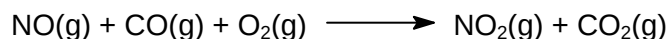
Ion	D ²⁺	E ⁺	F ⁺	G ²⁻	H ⁻	J ⁻
Radius/nm	0.16	0.19	0.15	0.16	0.19	0.15

DG, EH and FJ are ionic solids of the same lattice structure.

Which one of the following gives the correct order of their lattice energies with the lowest numerical value first?

A	EH	FJ	DG
B	DG	EH	FJ
C	DG	FJ	EH
D	EH	DG	FJ

- 9 Nitrogen monoxide in car exhaust is oxidised to nitrogen dioxide by carbon monoxide and oxygen.



Measurements of the initial rates of reaction give the relative results shown.

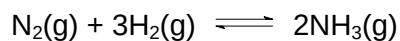
experiment	[NO(g)]	[CO(g)]	[O ₂ (g)]	initial rate
1	1	1	1	1
2	2	1	1	4
3	2	2	1	4
4	4	1	2	16

What are the orders of the reaction with respect to the three reactants?

	NO(g)	CO(g)	O ₂ (g)
A	1	0	1
B	2	0	0
C	2	1	0
D	2	1	1

- 10 When 7 mol of nitrogen gas and 7 mol of hydrogen gas were put into a 2 dm³ container and heated, the equilibrium mixture contained 2 mol of ammonia gas.

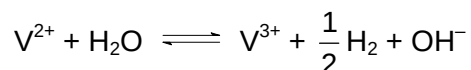
The equilibrium is represented by the following equation.



What is the numerical value of the equilibrium constant, K_c , at the temperature of the experiment?

- A** $\frac{1}{3(2^3)}$ **B** $\frac{2^2}{1(3^3)}$ **C** $\frac{2^2}{6(4^3)}$ **D** $\frac{2^2}{3(2 \times 3)^3}$

- 11 When vanadium(II) compounds are dissolved in water, the following equilibrium is established.

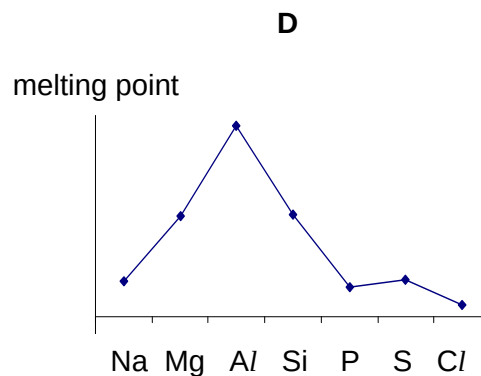
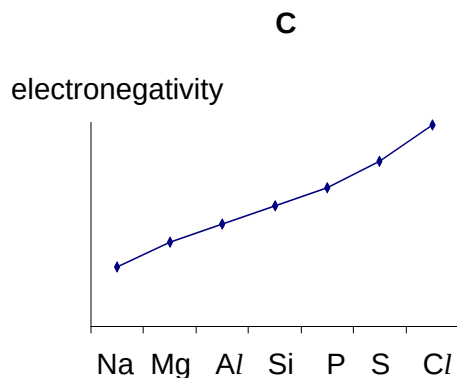
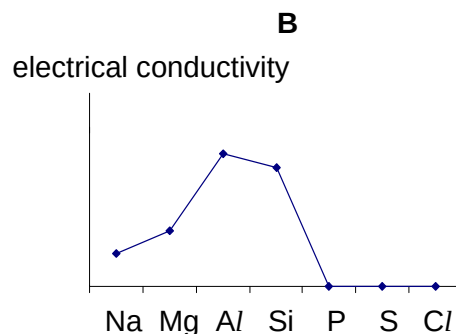
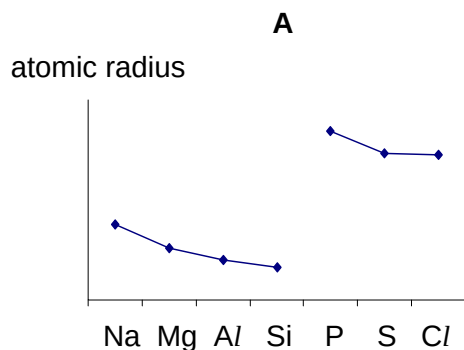


The percentage of V^{3+} present at equilibrium in two separate experiments at constant temperature is given below.

	Experiment 1	Experiment 2
Equilibrium percentage of V^{3+}	25	10

Which of the following statements is correct?

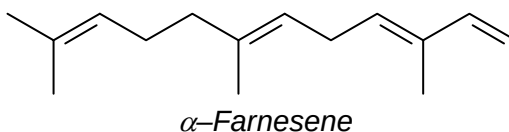
- A K_c decreased from experiment 1 to experiment 2.
 B In experiment 1, the solution was made less alkaline.
 C In experiment 2, the hydrogen was allowed to escape as it forms.
 D In experiment 2, a reagent that selectively precipitates V^{3+} ions was added.
- 12 What is the pH of the aqueous solution obtained by mixing 30 cm^3 of $0.1 \text{ mol dm}^{-3} \text{ HCl}$ and 10 cm^3 of $0.1 \text{ mol dm}^{-3} \text{ NaOH}$?
- A 1.30 B 1.18 C 1.12 D 1.00
- 13 Which of the following sketches shows the correct trend in the stated property, for the elements in the third period of the Periodic Table?



- 14 Which of the following decreases for the sequence of oxides, Na_2O , MgO and Al_2O_3 ?
- A melting point
 - B covalent character
 - C solubility in aqueous alkali
 - D pH of resultant solution when mixed with water
- 15 The oxide and chloride of an element **L** are separately added to water. The two resulting solutions have the same effect on litmus.

Which element could **L** be?

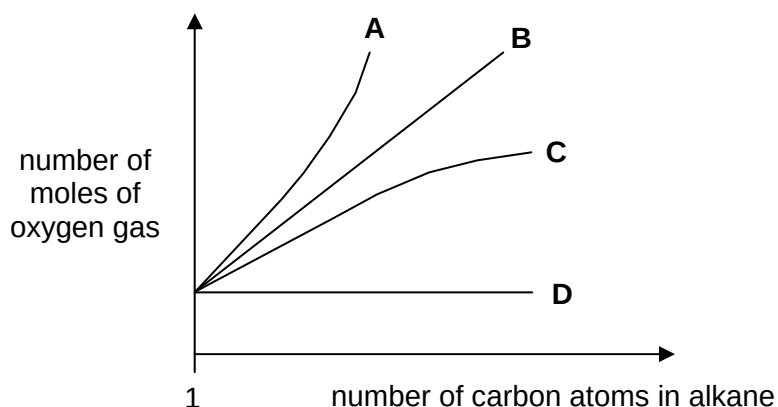
- A sodium
 - B magnesium
 - C silicon
 - D phosphorus
- 16 *α -Farnesene* is a constituent of the natural wax found on apples. It is also responsible for the characteristic odour of green apples.



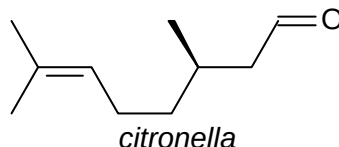
How many geometric isomers does this molecule have?

- A 2
 - B 4
 - C 8
 - D 16
- 17 The complete combustion of alkanes to produce carbon dioxide and water is an important exothermic reaction.

Which line on the graph shows the relationship between the number of carbon atoms in the alkane and the number of moles of oxygen gas needed for complete combustion of the alkane?



- 18** Natural muscone is obtained from the musk deer which has been used in perfumery and medicine for thousands of years. Since obtaining muscone requires the killing of the endangered animal, nearly all muscone used in perfumery are produced synthetically. One such synthesis uses *citronella*.



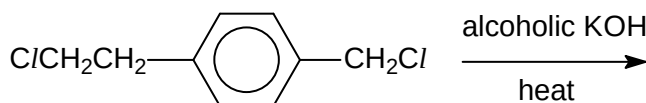
Which of the following statements about citronella is correct?

- A** It reacts with steam in the presence of H_3PO_4 to give a secondary alcohol.
- B** It gives black deposits of Ag when warmed with Fehling's solution.
- C** It reacts with hot, acidified KMnO_4 to give effervescence of CO_2 .
- D** It reacts with 2 mol of H_2 gas in the presence of platinum catalyst.
- 19** In a preparation of ethene, ethanol is added a drop at a time to a heated reagent **M**. To purify the ethene, it is bubbled through a solution **N** and then collected.

What could reagent **M** and solution **N** be?

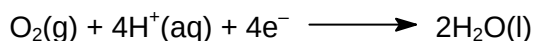
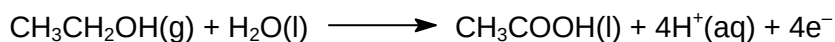
	reagent M	solution N
A	acidified $\text{K}_2\text{Cr}_2\text{O}_7$	dilute NaOH
B	concentrated H_2SO_4	dilute H_2SO_4
C	concentrated H_2SO_4	dilute NaOH
D	ethanolic NaOH	concentrated H_2SO_4

- 20** What is the expected product of the following reaction?



- A** HOCH_2CH_2 -C₆H₄-CH₂Cl
- B** HOCH_2CH_2 -C₆H₄-CH₂OH
- C** $\text{CH}_2=\text{CH}$ -C₆H₄-CH₂Cl
- D** $\text{CH}_2=\text{CH}$ -C₆H₄-CH₂OH

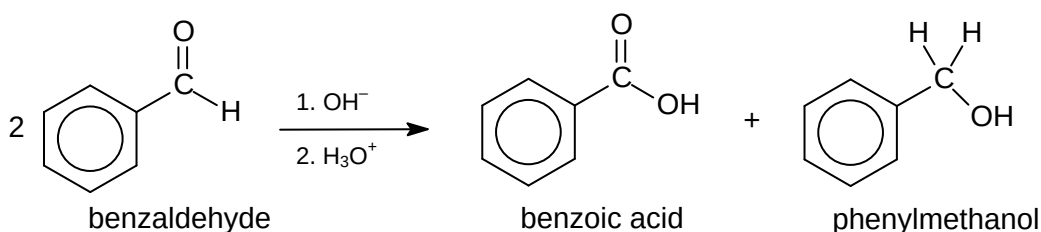
- 21 A breathalyzer is a device for estimating blood alcohol content from a breath sample. It is often used by traffic police at road blocks to identify drivers who are suspected of drink driving. When a suspected drink driver exhales into the breathalyzer, the following half reactions take place.



The electron transfer produces an electric current. The electric current is measured by a microprocessor and displayed as an approximation of overall blood alcohol content.

Which of the following statement is **not** true?

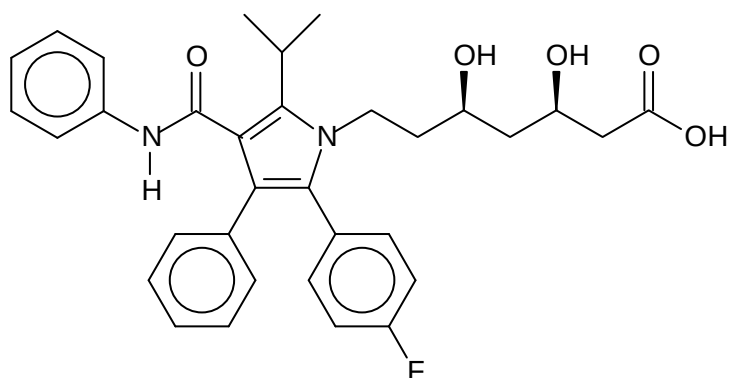
- A 2 mol of water is produced from 1 mol of ethanol.
 - B Propanol will produce the same electric current as ethanol.
 - C Ethanol will give the same products in the presence $\text{K}_2\text{Cr}_2\text{O}_7$.
 - D The two carbon atoms in ethanoic acid have different oxidation numbers.
- 22 The Cannizzaro reaction, discovered in 1853 by Stanislao Cannizzaro, is a common laboratory analogy for an important biological pathway by which reductions occur in living organisms. In this reaction, benzaldehyde gives two products, benzoic acid and phenylmethanol.



What is the type of reaction undergone by benzaldehyde?

- A acid–base
- B addition
- C disproportionation
- D substitution

23 *Atorvastatin* is a drug used to lower the blood cholesterol.



Atorvastatin

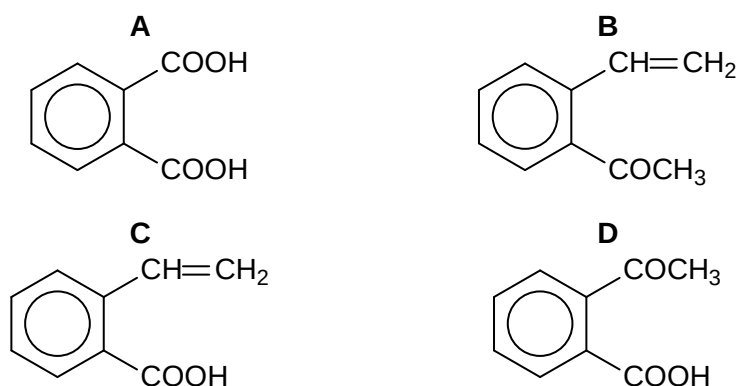
Which of the following statements about the molecule is true?

- A** It gives a bright orange precipitate when reacted with 2,4-DNPH.
- B** It reacts with one mole of Na metal to give three moles of H_2 gas.
- C** It reacts with $LiAlH_4$ to give a product that produces a yellow precipitate when warmed with alkaline aqueous iodine.
- D** It reacts with anhydrous PCl_5 to give white fumes.

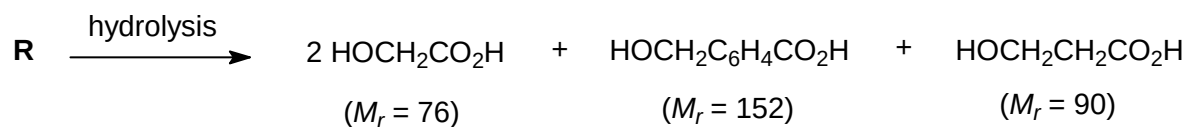
24 An organic compound **P** has the following properties:

- (i) It has molecular formula $C_9H_{10}O_2$.
- (ii) It gives a silver mirror with Tollens' reagent.
- (iii) It gives yellow crystals with alkaline aqueous iodine.
- (iv) Heating **P** with Al_2O_3 at $300\text{ }^\circ\text{C}$, produces a compound **Q** which gives an effervescence with hot acidified $KMnO_4$.

Which is the structure of the compound formed by oxidation of **Q** by acidified $K_2Cr_2O_7$?



- 25 Polyester is a polymer which contains the ester functional group. A small polyester **R** is hydrolysed according to the following reaction.



What is the M_r of **R**?

- A** 318 **B** 322 **C** 340 **D** 394

Section B

For each of the question in this section, one or more of the three numbered statements 1 to 3 may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct.)

The responses **A** to **D** should be selected on the basis of

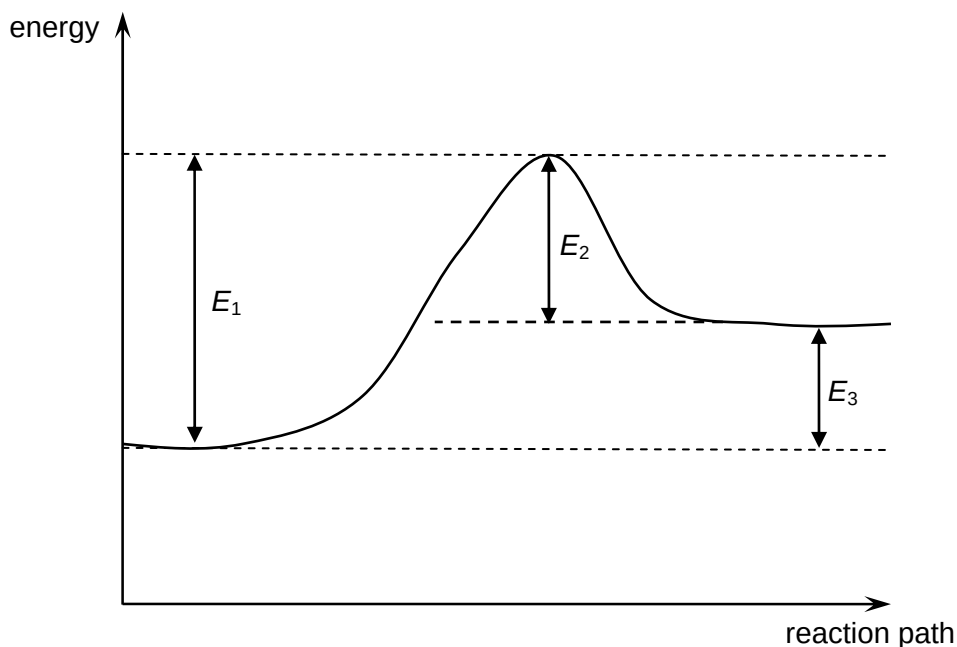
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

26 Which of the following property is a result of intermolecular hydrogen bonding?

- 1 Ice has a lower density than water at 0 °C.
- 2 The bond angle about the oxygen atom in ice is 109.5°.
- 3 The relative molecular mass of ethanoic acid, when dissolved in benzene, is 120.

27 The energy profile for a reversible reaction is shown below.

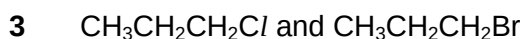
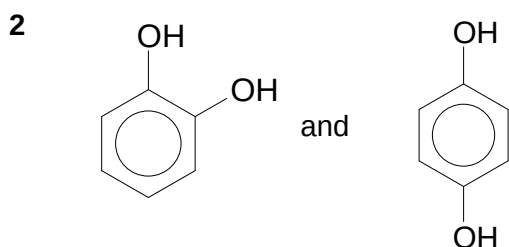
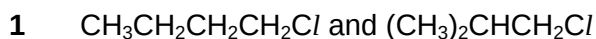


From the diagram, it can be deduced that

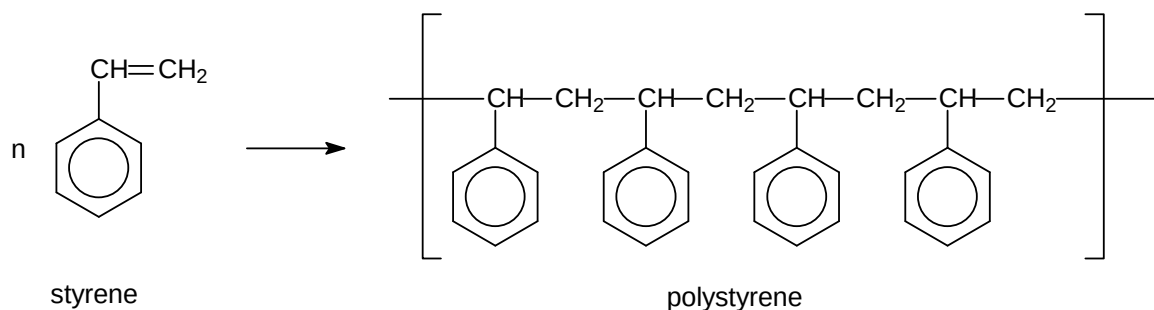
- 1 the reverse reaction is exothermic.
- 2 the addition of a catalyst changes the value of E_1 but not E_2 .
- 3 increasing the temperature increases the value of E_3 .

- 28 Which of the statements are correct about magnesium, sulfur and their oxides?
- 1 The number of moles of oxygen gas needed to completely react with 1 mol of magnesium is less than that with 1 mol of sulfur.
 - 2 The melting point of magnesium oxide is higher than that of sulfur dioxide.
 - 3 The pH of a solution formed from dissolving magnesium oxide in water is higher than that of sulfur dioxide.

- 29 In which of the following pairs does the first compound have a higher boiling point?



- 30 Polystyrene is a macromolecule commonly used in manufacturing disposable cups and food containers. It can be produced industrially by “joining up” many styrene molecules.



Which statements are correct?

- 1 Styrene decolourises aqueous bromine but polystyrene does not.
- 2 The conversion of styrene molecules into polystyrene involves addition process.
- 3 All atoms lie on the same plane for both styrene and polystyrene.

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	C	16	B
2	B	17	B
3	C	18	D
4	C	19	C
5	B	20	C
6	B	21	A
7	D	22	C
8	A	23	D
9	B	24	C
10	A	25	C
11	B	26	A
12	A	27	D
13	C	28	A
14	D	29	D
15	D	30	B